

Project Planning & Timeline

1. Project Phases & Milestones

The development of Opti-Crop was executed over a 6-week timeline, divided into focused sprint deliverables.

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- W1: Brainstorming & Requirements
- W2: Data Cleaning & ML Training
- W3: Database Setup & Backend Development
- W4: UI Implementation & Styling
- W5: Integration & Functional Testing
- W6: Production Deployment & Documentation

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2. Task Allocation

- **Ideation & Requirement Analysis (Week 1):** Scoped user needs, drafted DFDs, compiled solution functional specifications.
- **Data Wrangling & Model Engineering (Week 2):** Preprocessed crop recommendation data from Kaggle, handled phosphorus outliers, trained Logistic Regression classifier, and extracted cluster insights.
- **Database & Business Logic Integration (Week 3):** Developed relational tables, integrated models with Flask web controllers, and implemented user session hashing.
- **Front-End Styling & UI Polish (Week 4):** Created a premium glassmorphic visual layout, styled dashboards, and ensured responsive grid layouts.
- **Testing & Quality Assurance (Week 5):** Conducted performance checks, load tests, and verified prediction accuracy metrics.
- **Deployment & Documentation (Week 6):** Relocated files into structured directories, verified Render build commands, and generated documentation guides.